Data-Backed Journal-Style Replication of *An Organizational Perspective on Patenting and Open Innovation*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of An Organizational Perspective on Patenting and Open Innovation using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 1,958 observations across 189 entities, and the batch headline estimate is 0.457 for Post-1989 Electronics Shift in Patenting with p = 0.001. The local analogue reproduces the directional pattern emphasized in the source paper: the post-1989 electronics-semiconductor interaction raises patenting and lowers external spillovers. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: patenting; open innovation; spillovers; electronics; pro-patent practices; local calculated analogue

1 Introduction

This replication begins from a paper-first reading of An Organizational Perspective on Patenting and Open Innovation and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks why some firms embraced stronger patenting and what that shift did to knowledge spillovers (Bhaskarabhatla and

Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The study is worth revisiting because The replication matters because debates about patenting and openness often rely on anecdotes rather than transparent side-by-side evidence on patenting and spillovers. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 1,958 observations across 189 entities and spans 1986 to 2000 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Bhaskarabhatla and Hegde, 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a clear sector-period analogue that recovers the joint movement in patenting and spillovers from processed local firm-year data. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Bhaskarabhatla and Hegde, 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper sits in a top-journal literature on intellectual property, openness, appropriability, and the organizational conditions under which patenting both protects value and constrains spillovers. (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the replication is explicitly comparative: it asks whether a local sector-period analogue can recover the joint rise in patenting and fall in spillovers associated with stronger pro-patent behavior. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search,

technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The paper-specific review also adds classic and top-journal work on appropriability, spillovers, and innovation metrics so the patenting-openness tradeoff is discussed inside a broader empirical tradition. (Jaffe, 1989; Griliches, 1990). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Hall et al., 2005; Fleming and Sorenson, 2004). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Lavie et al., 2011; Ghosh et al., 2014).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Bhaskarabhatla and Hegde, 2014). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper links stronger patenting to organizational change and pro-patent practices. The consequences include both private performance and spillover effects. IBM and the electronics-semiconductors environment are central to the source argument. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Bhaskarabhatla and Hegde, 2014; March,

1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Stronger patenting and reduced spillovers are both central outcomes. The local reconstruction therefore adopts the following bounded decision: The local rerun estimates patenting and spillovers jointly in a sector-period panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Electronics and semiconductors are central to the setting. The local reconstruction therefore adopts the following bounded decision: Electronics-semiconductor firms are identified with local SIC codes. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Organizational change matters in the source narrative. The local reconstruction therefore adopts the following bounded decision: Firm-internal organizational change is translated into a transparent post-1989 sector interaction. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose

is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

DirectSourceFact	LocalReplicationDecision
Stronger patenting and reduced spillovers are both central outcomes.	The local rerun estimates patenting and spillovers jointly in a sector-period panel.
Electronics and semiconductors are central to the setting.	Electronics-semiconductor firms are identified with local SIC codes.
Organizational change matters in the source narrative.	Firm-internal organizational change is translated into a transparent post-1989 sector interaction.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Processed firm-year patent and publication counts from the current local public-firm panel. Aggregated corporate NPL-related counts that proxy outward knowledge spillovers. Financial controls and SIC-based sector coding from the local Compustat-linked panel (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 1,958 observations across 189 entities and covers 1986 to 2000 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those

verified evidence layers rather than to a hand-copied number from the article (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue reconstructs a post-1989 electronics-semiconductor shift in patenting and spillovers using processed firm-year patent, publication, and NPL-related counts from the current workspace. The local sample frame is Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding The observed time window in the executable panel is 1986 to 2000 in the local executable panel The outcome is Annual patent count and annual external corporate spillover count (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Bhaskarabhatla and Hegde, 2014; Hall et al., 2005).

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Bhaskarabhatla and Hegde (2014). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Bhaskarabhatla and Hegde, 2014;

Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Stage	NeededFor	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Yes	Direct source

Stage	NeededFor	Exists	Status
Analysis panel	Estimate the local model and regenerate descriptive statistics	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	Yes	Executable local result file
Fit summary	Check model diagnostics when a fit file is available	Yes	Executable local result file

Verified evidence layers used to support the paper's calculations.

3.2 Sample Reconstruction

The local analysis file contains 1,958 observations drawn from 189 entities. The observed time span runs from 1986 to 2000. Selected sample moments are Patents = 125.3, Publications = 107.8, External Spillovers = 11.99, Internal Spillovers = 5.235 (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	1,958
Unique Firms	189
Start Year	1986

Statistic	Value
End Year	2000
Share Electronics	14.35%
Mean Patents	125
Mean Publications	107.8
Mean External Spillovers	11.99
Mean Internal Spillovers	5.235

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Pro-patent shift is operationalized locally as Post-1989 electronics interaction and appears in the executable dataset as electronics_post. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Multiply the electronics-semiconductor indicator by a post-1989 time break.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Patenting response is operationalized locally as Annual patent count and appears in the executable dataset as pat_yr. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use processed annual patent counts.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the

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financial controls.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Pro-patent shift	Post-1989 electronics interaction	Multiply the electronics-semiconductor indicator by a post-1989 time break.
Patenting response	Annual patent count	Use processed annual patent counts.
Knowledge spillovers	External corporate spillover count	Use the processed external corporate spillover count.
Sector environment	Electronics-semiconductor indicator	Classify firms from SIC-based electronics and semiconductors codes.
Firm-year controls	Publication stock, patent stock, assets, R&D intensity	Reuse the local stock and financial controls.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Bhaskarabhatla and Hegde (2014). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Pro-patent shift, the manuscript uses Post-1989 electronics interaction under the exact local implementation electronics_post. The derivation step is: Multiply the electronics-

semiconductor indicator by a post-1989 time break.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patenting response, the manuscript uses Annual patent count under the exact local implementation pat_yr. The derivation step is: Use processed annual patent counts.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Knowledge spillovers, the manuscript uses External corporate spillover count under the exact local implementation corp_exter_cites. The derivation step is: Use the processed external corporate spillover count.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Sector environment, the manuscript uses Electronics-semiconductor indicator under the exact local implementation electronics_semiconductor. The derivation step is: Classify firms from SIC-based electronics and semiconductors codes.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm-year controls, the manuscript uses Publication stock, patent stock, assets, R&D intensity under the exact local implementation log_pub_stock; log_pat_stock; log_assets; rd_intensity. The derivation step is: Reuse the local stock and financial controls.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Patents per year has mean 125.3, standard deviation 218.8, and an interquartile range from 14.00 to 135.2; External spillovers has mean 11.99, standard deviation 52.69, and an interquartile range from 0.00000 to 5.000; Internal spillovers has mean 5.235, standard deviation 22.16, and an interquartile range from 0.00000 to 1.000; Post-1989 electronics shift has mean 0.115, standard deviation 0.319, and an interquartile range from 0.00000 to 0.00000 (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time-series

view is equally informative. FirmYears changes from 154.0 in the first observed period to 114.0 in the last observed period; MeanPatents changes from 84.81 in the first observed period to 181.9 in the last observed period; MeanExternalSpillovers changes from 1.571 in the first observed period to 37.21 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The grouped descriptive analysis is retained to show how the focal outcome behaves across the observed support of the main regressor (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Patents per year has missing share 0.00%; External spillovers has missing share 0.00%; Internal spillovers has missing share 0.00% . At the other end of the support, Patents per year remains close to complete with missing share 0.00%; External spillovers remains close to complete with missing share 0.00%; Internal spillovers remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. SUNOCO INC contributes 15 observations; TRW INC contributes 15 observations; TEXACO INC contributes 15 observations; TEXAS INSTRUMENTS INC contributes 15 observations; TEXTRON INC contributes 15 observations . The purpose of reporting

these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). A quantile-based support check is not available for this paper's current main regressor, so the discussion relies more heavily on grouped raw means and the descriptive distribution table (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because debates about patenting and openness often rely on anecdotes rather than transparent side-by-side evidence on patenting and spillovers. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005).

Variable	Missing	MissingShare
Patents per year	0	0.00%
External spillovers	0	0.00%
Internal spillovers	0	0.00%
Post-1989 electronics shift	0	0.00%
Electronics-semiconductor firm	0	0.00%
Log publication stock	0	0.00%
Log patent stock	0	0.00%
Log assets	0	0.00%
R&D intensity	0	0.00%

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Patents per year	1,958	125.3	218.8	14.00	57.00	135.2	0	2925
External spillovers	1,958	11.99	52.69	0.0000	0.0000	5.000	0	987
Internal spillovers	1,958	5.235	22.16	0.0000	0.0000	1.000	0	324

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Post-1989 electronics shift	1,958	0.115	0.319	0.0000	0.0000	0.0000	0	1
Electronics-semiconductor firm	1,958	0.144	0.351	0.0000	0.0000	0.0000	0	1
Log publication stock	1,958	4.641	2.237	3.069	4.689	6.562	0	9
Log patent stock	1,958	5.531	1.823	4.497	5.923	6.773	0	9
Log assets	1,958	8.692	1.288	7.934	8.600	9.402	4	13
R&D intensity	1,958	0.0402	0.0371	0.0114	0.0271	0.0592	0	0

Main descriptive statistics discussed in the body of the paper.

Period	FirmYears	MeanPatents	MeanExternalSpillovers	MeanPublications	ElectronicsShare
1986	154	85	1.571	85.54	12.34%
1987	147	100	2.116	91.83	12.93%
1988	137	95	2.540	99.51	13.14%
1989	132	117	5.439	91.00	13.64%
1990	135	107	4.815	96.80	13.33%
1991	133	118	5.940	103.4	13.53%
1992	134	116	6.612	106.6	13.43%
1993	132	121	8.174	117.5	13.64%
1994	128	128	9.898	130.8	14.06%
1995	125	128	10.92	132.1	14.40%
1996	126	136	15.02	112.1	15.87%
1997	124	130	21.51	113.9	16.94%
1998	121	169	28.02	111.8	17.36%
1999	116	179	31.31	109.8	15.52%
2000	114	182	37.21	125.7	16.67%

Observed time patterns in the current dataset.

Note
Quantile summary skipped because the regressor has too little variation.

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Patents per year	External spillovers	Internal spillovers	Post-1989 electronics shift	Electronics-semiconductor firm	Log publication stock	Log patent stock	Log assets	R&D intensity
Patents per year	1	0.569	0.649	0.174	0.153	0.526	0.580	0.544	0.175
External spillovers	1	1.000	0.752	0.036	0.019	0.317	0.253	0.296	0.141
Internal spillovers	1	0.752	1.000	0.015	-0.000	0.372	0.278	0.301	0.172
Post-1989 electronics shift	0	0.036	0.015	1.000	0.880	0.129	0.150	-0.032	0.266
Electronics-semiconductor firm	0	0.019	-0.000	0.880	1.000	0.140	0.149	-0.080	0.305
Log publication stock	1	0.317	0.372	0.129	0.140	1.000	0.749	0.563	0.374

Compact correlation structure for the main variables.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is Firm and year fixed-effects count models for patenting and external spillovers with a post-1989 electronics-semiconductor interaction. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Bhaskarabhatla and Hegde, 2014; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame

is Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding, and the empirical window is 1986 to 2000 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Annual patent count and annual external corporate spillover count, and the main explanatory design is Firm and year fixed-effects count models with a post-1989 electronics-semiconductor interaction. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Publication stock, patent stock where relevant, assets, and R&D intensity plus firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps firm-years with verified sector coding and processed patent-publication measures and uses a sector-period interaction when firm-internal practice changes are not directly observed.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

DecisionPoint	LocalSpecification
Research question	The source paper asks why some firms embraced stronger patenting and what that shift did to knowledge spillovers.

DecisionPoint	LocalSpecification
Unit of analysis	Firm-year
Sample frame	Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding.
Time window	1986 to 2000 in the local executable panel.
Outcome definition	Annual patent count and annual external corporate spillover count.
Main explanatory design	Firm and year fixed-effects count models with a post-1989 electronics-semiconductor interaction.
Controls and fixed effects	Publication stock, patent stock where relevant, assets, and R&D intensity plus firm and year fixed effects.
Exclusion or substitution logic	The local rerun keeps firm-years with verified sector coding and processed patent-publication measures and uses a sector-period interaction when firm-internal practice changes are not directly observed.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding over 1986 to 2000 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Annual patent count and annual external corporate spillover count and the main explanatory design is Firm and year fixed-effects count models with a post-1989 electronics-semiconductor interaction. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Publication stock, patent stock where relevant, assets, and R&D intensity plus firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps firm-years with verified sector coding and processed patent-publication measures and uses a sector-period interaction when firm-internal practice changes are not directly observed. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$E[Y_{it} \mid X_{it}] = \exp\left(\beta_1(\text{Electronics}_i \times \text{Post1989}_t) + \beta_2 X_{it} + \alpha_i + \gamma_t\right). \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.457 on Post-1989 Electronics Shift in Patenting (p = 0.001). The local analogue reproduces the directional pattern emphasized in the source paper: the post-1989 electronics-semiconductor interaction raises patenting and lowers external spillovers. Log publication stock is estimated at 0.286 with p = ¡ .001. Post-1989 electronics shift is estimated at 0.457 with p = 0.001. Log patent stock is estimated at 0.425 with p = 0.007 (Bhaskarabhatla and Hegde,

2014; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Model	Term	Estimate	Std. error	P-value
Patenting	Post-1989 electronics shift	0.4569	0.1401	0.0011
Patenting	Log publication stock	0.2861	0.0774	0.0002
Patenting	Log assets	0.1631	0.1733	0.347
Patenting	R&D intensity	-1.515	2.522	0.548
Spillovers	Post-1989 electronics shift	-0.5009	0.2352	0.0332
Spillovers	Log publication stock	-0.0837	0.1003	0.404
Spillovers	Log patent stock	0.4251	0.1585	0.0073
Spillovers	Log assets	0.0582	0.1607	0.717
Spillovers	R&D intensity	4.489	2.133	0.0354

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log publication stock is estimated at 0.286 with a p-value of ¡ .001; Post-1989 electronics shift is estimated at 0.457 with a p-value of 0.001; Log patent stock is estimated at 0.425 with a p-value of 0.007; Post-1989 electronics shift is estimated at -0.501 with a p-value of 0.033 (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Log publication stock, which is estimated at 0.286 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Patents per year has mean 125.3

and standard deviation 218.8 in the local panel, while the main regressor Post-1989 electronics shift has mean 0.115 and standard deviation 0.319 (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The grouped summaries retained in the appendix are intended to show the raw outcome pattern underlying the formal specification (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue reproduces the directional pattern emphasized in the source paper: the post-1989 electronics-semiconductor interaction raises patenting and lowers external spillovers. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within patenting, the leading coefficient family reads as follows: Log publication stock = 0.286 (p = ¡ .001); Post-1989 electronics shift = 0.457 (p = 0.001); Log assets = 0.163 (p = 0.35); R&D intensity = -1.515 (p = 0.55). The largest positive estimate in this specification is Post-1989 electronics shift at 0.457, while the largest negative estimate is R&D intensity at -1.515. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within spillovers, the leading co-

efficient family reads as follows: Log patent stock = 0.425 (p = 0.007); Post-1989 electronics shift = -0.501 (p = 0.033); R&D intensity = 4.489 (p = 0.035); Log publication stock = -0.0837 (p = 0.40); Log assets = 0.0582 (p = 0.72). The largest positive estimate in this specification is R&D intensity at 4.489, while the largest negative estimate is Post-1989 electronics shift at -0.501. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Bhaskarabhatla and Hegde (2014) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Patenting, the most informative focal term is Log publication stock with estimate 0.286 and p = ¡ .001; In Spillovers, the most informative focal term is Log patent stock with estimate 0.425 and p = 0.007 (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Patenting specification contributes its own reading to the replication. Log publication stock is estimated at 0.286 with p = ¡ .001 and is classified as a strong signal; Post-1989 electronics shift is estimated at 0.457 with p = 0.001 and is classified as a strong signal; Log assets is estimated at 0.163 with p = 0.35 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Spillovers specification contributes its own reading to the replication. Log patent stock is estimated at 0.425 with p = 0.007 and is classified as a strong signal; Post-1989 electronics shift is estimated at -0.501 with p = 0.033 and is classified

as a moderate signal; R&D intensity is estimated at 4.489 with p = 0.035 and is classified as a moderate signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Bhaskarabhatla and Hegde (2014) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Term	Model	Estimate	PValue	Signal	Reading
Post-1989 electronics shift	Patenting	0.4569	0.001	Strong	Positive estimate with p = 0.001.
Log publication stock	Patenting	0.2861	¡ .001	Strong	Positive estimate with p = ¡ .001.
Log assets	Patenting	0.1631	0.35	Weak	Positive estimate with p = 0.35.
R&D intensity	Patenting	-1.515	0.55	Weak	Negative estimate with p = 0.55.
Post-1989 electronics shift	Spillovers	-0.5009	0.033	Moderate	Negative estimate with p = 0.033.
Log publication stock	Spillovers	-0.0837	0.40	Weak	Negative estimate with p = 0.40.
Log patent stock	Spillovers	0.4251	0.007	Strong	Positive estimate with p = 0.007.

Term	Model	Estimate	PValue	Signal	Reading
Log assets	Spillovers	0.0582	0.72	Weak	Positive estimate with p = 0.72.
R&D intensity	Spillovers	4.489	0.035	Moderate	Positive estimate with p = 0.035.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). For patenting, the current fit export reports N = 1,910, pseudo R-squared = 0.902 (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). For spillovers, the current fit export reports N = 1,484, pseudo R-squared = 0.916 (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Bhaskarabhatla and Hegde (2014) is being carried out on a clearly described estimating sample with observable fit statistics (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005).

model	n	logLik	AIC	BIC	pseudo_r2
patenting	1,910	-20,116	40,600	41,622	0.902
spillovers	1,484	-3,867	7,993	8,677	0.916

Model fit statistics discussed in the main results section.

Model	Term	Estimate	Std. error	Statistic	P-value
Patenting	Post-1989 electronics shift	0.4569	0.1401	3.262	0.0011
Patenting	Log publication stock	0.2861	0.0774	3.696	0.0002

Model	Term	Estimate	Std. error	Statistic	P-value
Patenting	Log assets	0.1631	0.1733	0.941	0.347
Patenting	R&D intensity	-1.515	2.522	-0.601	0.548
Spillovers	Post-1989 electronics shift	-0.5009	0.2352	-2.130	0.0332
Spillovers	Log publication stock	-0.0837	0.1003	-0.834	0.404
Spillovers	Log patent stock	0.4251	0.1585	2.682	0.0073
Spillovers	Log assets	0.0582	0.1607	0.362	0.717
Spillovers	R&D intensity	4.489	2.133	2.104	0.0354

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Bhaskarabhatla and Hegde, 2014). For Bhaskarabhatla and Hegde (2014), the local headline result is Post-1989 Electronics Shift in Patenting = 0.457 with p = 0.001. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper argues that stronger pro-patent practices increased patenting and licensing but curtailed knowledge spillovers.	Post-1989 Electronics Shift in Patenting = 0.457 with p = 0.001.	The current local panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break.
Measurement and data construction	The published design combines detailed organizational evidence with firm-level patenting and spillover consequences.	The local analogue reconstructs a post-1989 electronics-semiconductor shift in patenting and spillovers using processed firm-year patent, publication, and NPL-related counts from the current workspace.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published analysis focuses on IBM and related large U.S. corporations in electronics and semiconductors.	Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding. Local sample = 1,958 observations across 189 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design combines detailed organizational evidence with firm-level patenting and spillover consequences By contrast, the local paper relies on the following executable design: The local analogue reconstructs a post-1989 electronics-semiconductor shift in patenting and spillovers using processed firm-year patent, publication, and NPL-related counts from the current workspace These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published analysis focuses on IBM and related large U.S. corporations in electronics and semiconductors The local sample instead works from Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding and yields 1,958 observations across 189 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The current local panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March,

1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The current local panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The current local panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The source paper studies firm-specific organizational change and leadership in detail, especially at IBM (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns

sample translation. The local rerun does not observe managerial practice adoption directly and instead uses a sector-by-period analogue (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The exercise therefore targets the directional consequences of pro-patent behavior rather than the internal narrative of one firm (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a clear sector-period analogue that recovers the joint movement in patenting and spillovers from processed local firm-year data (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Bhaskarabhatla and Hegde, 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	1,958	rows
Unique Firms	189	unique_firms
Start Year	1986	year_min
End Year	2000	year_max
Share Electronics	14.35%	share_electronics
Mean Patents	125	mean_patents
Mean Publications	107.8	mean_publications
Mean External Spillovers	11.99	mean_external_spillovers
Mean Internal Spillovers	5.235	mean_internal_spillovers

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Patents per year	1,958	125.3	218.8	14.00	57.00	135.2	0	2925
External spillovers	1,958	11.99	52.69	0.0000	0.0000	5.000	0	987
Internal spillovers	1,958	5.235	22.16	0.0000	0.0000	1.000	0	324
Post-1989 electronics shift	1,958	0.115	0.319	0.0000	0.0000	0.0000	0	1
Electronics-semiconductor firm	1,958	0.144	0.351	0.0000	0.0000	0.0000	0	1
Log publication stock	1,958	4.641	2.237	3.069	4.689	6.562	0	9
Log patent stock	1,958	5.531	1.823	4.497	5.923	6.773	0	9
Log assets	1,958	8.692	1.288	7.934	8.600	9.402	4	13
R&D intensity	1,958	0.0402	0.0371	0.0114	0.0271	0.0592	0	0

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Patents per year	External spillovers	Internal spillovers	Post-1989 electronics shift	Electronics-semiconductor firm	Log publication stock	Log patent stock	Log assets	R&D intensity
Patents per year	1	0.569	0.649	0.174	0.153	0.526	0.580	0.544	0.175
External spillovers	1	1.000	0.752	0.036	0.019	0.317	0.253	0.296	0.141
Internal spillovers	1	0.752	1.000	0.015	-0.000	0.372	0.278	0.301	0.172
Post-1989 electronics shift	0	0.036	0.015	1.000	0.880	0.129	0.150	-0.032	0.266
Electronics-semiconductor firm	0	0.019	-0.000	0.880	1.000	0.140	0.149	-0.080	0.305
Log publication stock	1	0.317	0.372	0.129	0.140	1.000	0.749	0.563	0.374
Log patent stock	1	0.253	0.278	0.150	0.149	0.749	1.000	0.581	0.189
Log assets	1	0.296	0.301	-0.032	-0.080	0.563	0.581	1.000	-0.062
R&D intensity	0	0.141	0.172	0.266	0.305	0.374	0.189	-0.062	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	FirmYears	MeanPatents	MeanExternalSpillovers	MeanPublications	ElectronicsShare
1986	154	85	1.571	85.54	12.34%
1987	147	100	2.116	91.83	12.93%
1988	137	95	2.540	99.51	13.14%
1989	132	117	5.439	91.00	13.64%
1990	135	107	4.815	96.80	13.33%
1991	133	118	5.940	103.4	13.53%
1992	134	116	6.612	106.6	13.43%
1993	132	121	8.174	117.5	13.64%
1994	128	128	9.898	130.8	14.06%
1995	125	128	10.92	132.1	14.40%
1996	126	136	15.02	112.1	15.87%
1997	124	130	21.51	113.9	16.94%
1998	121	169	28.02	111.8	17.36%
1999	116	179	31.31	109.8	15.52%
2000	114	182	37.21	125.7	16.67%

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Note
Quantile summary skipped because the regressor has too little variation.

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

model	term	estimate	std_error	statistic	p_value
patenting	electronics_post	0.4569	0.1401	3.262	0.001
patenting	log_pub_stock	0.2861	0.0774	3.696	¡ .001
patenting	log_assets	0.1631	0.1733	0.941	0.35
patenting	rd_intensity	-1.515	2.522	-0.601	0.55
spillovers	electronics_post	-0.5009	0.2352	-2.130	0.033
spillovers	log_pub_stock	-0.0837	0.1003	-0.834	0.40
spillovers	log_pat_stock	0.4251	0.1585	2.682	0.007
spillovers	log_assets	0.0582	0.1607	0.362	0.72
spillovers	rd_intensity	4.489	2.133	2.104	0.035

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Patents per year	0	0.00%
External spillovers	0	0.00%
Internal spillovers	0	0.00%
Post-1989 electronics shift	0	0.00%
Electronics-semiconductor firm	0	0.00%
Log publication stock	0	0.00%
Log patent stock	0	0.00%
Log assets	0	0.00%
R&D intensity	0	0.00%

Missingness audit for the variables summarized in the manuscript.

Entity	Label	Observations	FirstPeriod	LastPeriod
10156	SUNOCO INC	15	1986	2000
10301	TRW INC	15	1986	2000
10482	TEXACO INC	15	1986	2000
10499	TEXAS INSTRUMENTS INC	15	1986	2000
10519	TEXTRON INC	15	1986	2000
10581	TIMKEN CO	15	1986	2000
1078	ABBOTT LABORATORIES	15	1986	2000
10857	UNION CARBIDE CORP	15	1986	2000
10983	RTX CORP	15	1986	2000
11465	WHIRLPOOL CORP	15	1986	2000

Entity	Label	Observations	FirstPeriod	LastPeriod
1300	HONEYWELL INTERNATIONAL INC	15	1986	2000
1356	ALCOA INC	15	1986	2000
1408	BEAM INC	15	1986	2000
1478	WYETH	15	1986	2000
1581	AT&T CORP	15	1986	2000
1651	ANDREW CORP	15	1986	2000
1690	APPLE INC	15	1986	2000
1722	ARCHER-DANIELS-MIDLAND CO	15	1986	2000
1762	ARMSTRONG HOLDINGS INC	15	1986	2000
2085	BAUSCH & LOMB HLDGS -REDH	15	1986	2000

Top entities by local observation count.

References

Ahuja, G. (2000). Collaboration networks, structural holes, and innovation: A longitudinal study. *Administrative Science Quarterly*, 45(3):425–455.

Ahuja, G. and Katila, R. (2001). Technological acquisitions and the innovation performance of acquiring firms: A longitudinal study. *Strategic Management Journal*, 22(3):197–220.

Ahuja, G. and Lampert, C. M. (2001). Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22(6-7):521–543.

Almeida, P. and Kogut, B. (1999). Localization of knowledge and the mobility of engineers in regional networks. *Management Science*, 45(7):905–917.

Anderson, P. and Tushman, M. L. (1986). Technological discontinuities and organizational environments. *Administrative Science Quarterly*, 31(3):439–465.

Benner, M. J. and Tushman, M. L. (2003). Exploitation, exploration, and process management: The productivity dilemma revisited. *Academy of Management Review*, 28(2):238–256.

Bhaskarabhatla, A. and Hegde, D. (2014). An organizational perspective on patenting and open innovation. *Organization Science*, 25(6):1744–1763.

Cohen, W. M. and Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1):128–152.

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1):117–132.

Fleming, L., Mingo, S., and Chen, D. (2007). Collaborative brokerage, generative creativity, and creative success. *Administrative Science Quarterly*, 52(3):443–475.

Fleming, L. and Sorenson, O. (2004). Science as a map in technological search. *Strategic Management Journal*, 25(8-9):909–928.

Ghosh, A., Martin, X., Pennings, J. M., and Wezel, F. C. (2014). Ambition is nothing without focus: Compensating for negative transfer of experience in r&d. *Organization Science*, 25(2):572–590.

Griliches, Z. (1990). Patent statistics as economic indicators: A survey. *Journal of Economic Literature*, 28(4):1661–1707.

Hall, B. H., Jaffe, A., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, 36(1):16–38.

Hargadon, A. and Sutton, R. I. (1997). Technology brokering and innovation in a product development firm. *Administrative Science Quarterly*, 42(4):716–749.

Henderson, R. and Cockburn, I. (1994). Measuring competence? exploring firm effects in pharmaceutical research. *RAND Journal of Economics*, 25(1):63–84.

Henderson, R. M. and Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35(1):9–30.

Jaffe, A. B. (1986). Technological opportunity and spillovers of r&d: Evidence from firms' patents, profits, and market value. *American Economic Review*, 76(5):984–1001.

Jaffe, A. B. (1989). Real effects of academic research. *American Economic Review*, 79(5):957–970.

Jaffe, A. B., Trajtenberg, M., and Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *Quarterly Journal of Economics*, 108(3):577–598.

Katila, R. and Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194.

Kogut, B. and Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3(3):383–397.

Kogut, B. and Zander, U. (1996). What firms do? coordination, identity, and learning. *Organization Science*, 7(5):502–518.

Lavie, D., Kang, J., and Rosenkopf, L. (2011). Balance within and across domains: The performance implications of exploration and exploitation in alliances. *Organization Science*, 22(6):1517–1538.

Levinthal, D. A. and March, J. G. (1993). The myopia of learning. *Strategic Management Journal*, 14(S2):95–112.

Levitt, B. and March, J. G. (1988). Organizational learning. *Annual Review of Sociology*, 14:319–338.

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87.

Rosenkopf, L. and Almeida, P. (2003). Overcoming local search through alliances and mobility. *Management Science*, 49(6):751–766.

Rosenkopf, L. and Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306.

Schilling, M. A. and Phelps, C. C. (2007). Interfirm collaboration networks and knowledge creation: The impact of large-scale network structure on firm innovation. *Management Science*, 53(7):1113–1126.

Stuart, T. E. (2000). Interorganizational alliances and the performance of firms: A study of growth and innovation rates in a high-technology industry. *Strategic Management Journal*, 21(8):791–811.

Stuart, T. E. and Podolny, J. M. (1996). Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(S1):21–38.

Teece, D. J., Pisano, G., and Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7):509–533.

Trajtenberg, M. (1990). A penny for your quotes: Patent citations and the value of innovations. *RAND Journal of Economics*, 21(1):172–187.

Tripsas, M. and Gavetti, G. (2000). Capabilities, cognition, and inertia: Evidence from digital imaging. *Strategic Management Journal*, 21(10-11):1147–1161.